The radiopharmaceutical 2-deoxy-2-(^{18}F)fluoro-D-glucose (abbreviated as ^{18}F-FDG) is a modified form of D-glucose in which the hydroxyl group on the 2-carbon of the molecule has been replaced with a positron-emitting ^{18}F radionuclide. Synthesis of ^{18}F-FDG is accomplished in several steps involving both nuclear and chemical reactions, as shown in Figure 1.

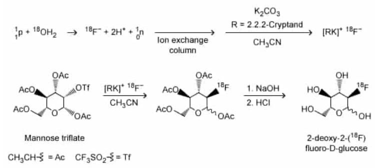


Figure 1 Synthetic scheme for 2-deoxy-2-(^{18}F)fluoro-D-glucose

Initially, a target of ^{18}O-enriched water ($^{18}OH_2$) is bombarded by protons using a cyclotron to achieve a (p-n) nuclear reaction in which a proton hitting the ^{18}O nucleus is absorbed by the nucleus, causing the ejection of a neutron and forming an ^{18}F nucleus. Following separation of the resulting $^{18}F^-$ anions from the residual $^{18}OH_2$ via an ion exchange column, $^{18}F^-$ is used in a nucleophilic substitution reaction with an acetylated mannose triflate substrate. Subsequent hydrolysis to remove the acetyl protecting groups yields ^{18}F-FDG.

Because ^{18}F-FDG is an analog of D-glucose, it is readily taken into cells by the same metabolic processes as D-glucose. Once inside the cells, the positron emissions of the ^{18}F radionuclide can be detected by positron emission tomography (PET) scans to produce diagnostic medical images of objects such as tumors. Emissions from the ^{18}F radionuclide subsequently diminish over time via radioactive decay (Figure 2).

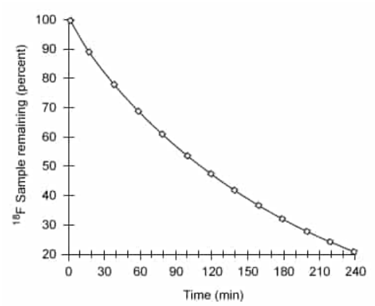


Figure 2 Nuclear decay of the ^{18}F radionuclide

Because of limitations in the spatial resolution of PET scanners, when an object (such as a tumor) is smaller than 30 mm in diameter, the measured specific uptake value (SUV) of ^{18}F-FDG accumulated within the object is lower than the actual SUV. The recovery coefficient (the ratio of the measured SUV to the actual SUV) shows that the measured SUV decreases as the size of the object decreases below the 30-mm threshold (Figure 3).

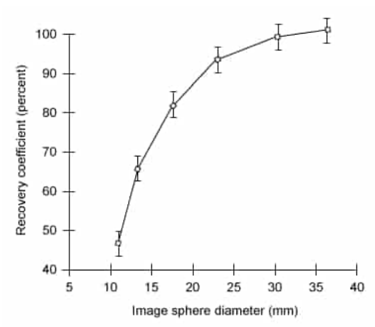
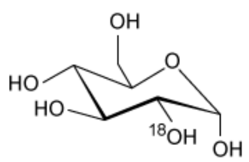


Figure 3 Recovery coefficient as a function of object size in a PET scan with a spatial resolution of 7 mm using ordered subsets expectation maximization (OSEM) image processing

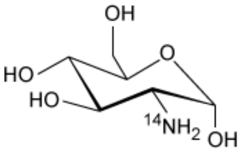
Which of the following products is formed from ^{18}F-FDG following a positron emission? (Note: Assume any anions produced acquire hydrogen ions from the aqueous physiological environment.)

A.



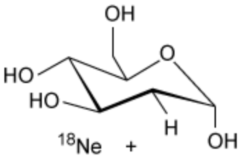
[46%]

◯ B.



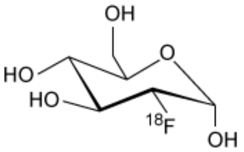
[6%]

◯ C.



[12%]

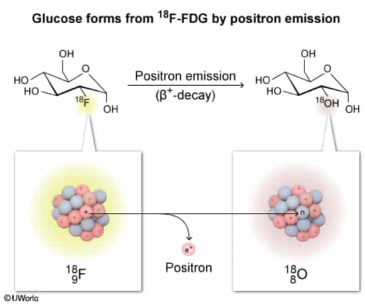
◯ D.



[35%]

Correct
46% answered correctly

Explanation:



Radioactive beta decay can occur in three forms: β^--decay (electron emission), β^+-decay (**positron emission**), and electron capture. In β^--decay, a neutron converts to a nuclear proton and emits an electron. In **β^+-decay**, a nuclear **proton converts** to a **neutron** (the opposite of β^--decay) and ejects a positron (an electron with a positive charge). In electron capture, a proton captures an electron near the nucleus and converts to a neutron without a positron or electron emission.

In all three forms of beta decay, the mass number remains **unchanged** whereas the atomic number either increases (β^--decay) or decreases (β^+-decay and electron capture) by 1. As the atomic number changes, the identity of the element changes accordingly.

As stated in the passage, ^{18}F is a positron-emitter (undergoes β^+-decay). When the ^{18}F in ^{18}F-FDG emits a positron (e^+) by β^+-decay, one of the protons in its nucleus is converted to a neutron, yielding a new nucleus with the same mass number (18) but an atomic number that is **decreased by 1** (9 − 1 = 8). The result is an ^{18}O nucleus.

$$^{18}_{9}\mathrm{F} \rightarrow {}^{18}_{8}\mathrm{O} + \mathrm{e}^{+}$$

Acquisition of a hydrogen ion by the oxygen anion yields a hydroxyl group appended to an inert glucose molecule.

(Choice B) A ^{14}N nucleus could form only if ^{18}F were to eject an alpha particle by alpha decay, but ^{18}F emits positrons by β^+-decay.

(Choice C) A ^{18}Ne nucleus would not form unless ^{18}F were to undergo β^--decay (electron emission) instead of β^+-decay (positron emission), but ^{18}F is a positron-emitter.

(Choice D) The mass number and identity of the ^{18}F nucleus would remain unchanged if it underwent a gamma decay, but ^{18}F undergoes β^+-decay.

Educational objective:
Positron emission occurs during β^+-decay, in which a nuclear proton converts to a neutron. This causes the atomic number of the nucleus to decrease by 1 while the mass number of the nucleus remains unchanged.

Time spent:0 sec
Subject:
General Chemistry

QID:402457
System:
Atomic Theory and Chemical Composition

The radiopharmaceutical 2-deoxy-2-(^{18}F)fluoro-D-glucose (abbreviated as ^{18}F-FDG) is a modified form of D-glucose in which the hydroxyl group on the 2-carbon of the molecule has been replaced with a positron-emitting ^{18}F radionuclide. Synthesis of ^{18}F-FDG is accomplished in several steps involving both nuclear and chemical reactions, as shown in Figure 1.

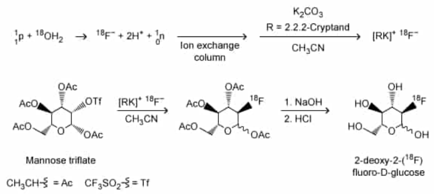


Figure 1 Synthetic scheme for 2-deoxy-2-(^{18}F)fluoro-D-glucose

Initially, a target of ^{18}O-enriched water ($^{18}OH_2$) is bombarded by protons using a cyclotron to achieve a (p-n) nuclear reaction in which a proton hitting the ^{18}O nucleus is absorbed by the nucleus, causing the ejection of a neutron and forming an ^{18}F nucleus. Following separation of the resulting $^{18}F^-$ anions from the residual $^{18}OH_2$ via an ion exchange column, $^{18}F^-$ is used in a nucleophilic substitution reaction with an acetylated mannose triflate substrate. Subsequent hydrolysis to remove the acetyl protecting groups yields ^{18}F-FDG.

Because ^{18}F-FDG is an analog of D-glucose, it is readily taken into cells by the same metabolic processes as D-glucose. Once inside the cells, the positron emissions of the ^{18}F radionuclide can be detected by positron emission tomography (PET) scans to produce diagnostic medical images of objects such as tumors. Emissions from the ^{18}F radionuclide subsequently diminish over time via radioactive decay (Figure 2).

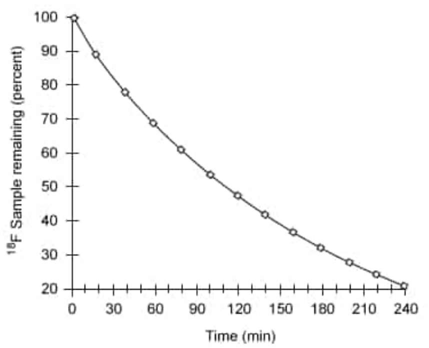


Figure 2 Nuclear decay of the ^{18}F radionuclide

Because of limitations in the spatial resolution of PET scanners, when an object (such as a tumor) is smaller than 30 mm in diameter, the measured specific uptake value (SUV) of ^{18}F-FDG accumulated within the object is lower than the actual SUV. The recovery coefficient (the ratio of the measured SUV to the actual SUV) shows that the measured SUV decreases as the size of the object decreases below the 30-mm threshold (Figure 3).

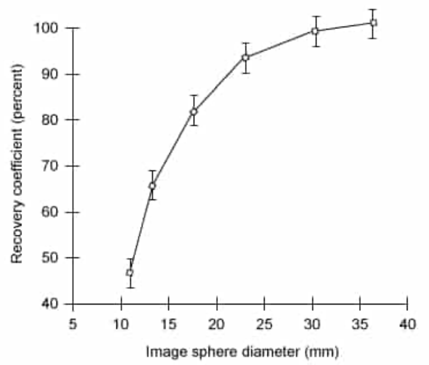


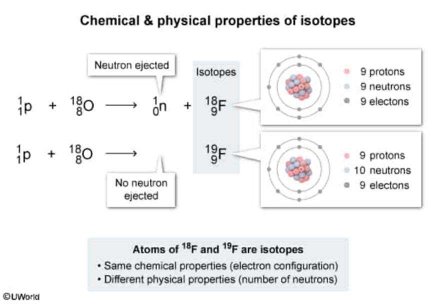
Figure 3 Recovery coefficient as a function of object size in a PET scan with a spatial resolution of 7 mm using ordered subsets expectation maximization (OSEM) image processing

Suppose that the (p-n) nuclear reaction described in the passage was attempted but resulted in the ^{18}O nucleus absorbing a proton without ejecting a neutron. Which of the following statements correctly describes the resulting atom that would be formed? (Assume any cations formed acquire missing valence electrons from the aqueous physiological environment.)

- A. The process would form an isotope of fluorine with very different chemical properties than ^{18}F. [25%]
- B. The process would form an isotope of oxygen with nearly identical chemical properties to ^{18}O. [10%]
- ✔ C. The process would form an isotope of fluorine with nearly identical chemical properties to ^{18}F. [59%]
- D. The process would form an element other than O and F with very different chemical properties. [5%]

Correct
59% answered correctly

Explanation:



Isotopes are atoms of the *same element* with a different number of neutrons in the nucleus. Isotopes are designated by their mass number (the sum of the **protons** and **neutrons** in the nucleus) either by using the element name hyphenated with its mass number (eg, fluorine-18), or by using a chemical symbol with the mass number given as a preceding superscript (eg, ^{18}F).

Because the atomic number (the **number of protons**) determines the *identity* of an element and is the same for all atoms of that element, the number of protons is not always explicitly stated in isotope notation. However, when stated, the atomic number is placed as a subscript preceding the chemical symbol and below the mass number.

The chemical behavior of an atom is determined primarily by its electron configuration, not by the number of neutrons in the nucleus. As a result, isotopes of the same element have nearly identical **chemical properties** (such as bonding and reactivity) but differ in their **physical properties** (such as density and mass).

If an ^{18}O nucleus were to absorb a proton, the atomic number would increase by 1 to a total of 9 protons. Because all atoms with 9 protons are fluorine atoms, the resulting nucleus would be a fluorine isotope. If no neutron was ejected following the absorption of the proton, the mass number would also increase by 1 to give a fluorine isotope with a mass number of 19. The fluorine cation initially formed would quickly acquire the missing valence electron from the surrounding environment.

$$^{18}_{8}\mathrm{O} + ^{1}_{1}\mathrm{p} \rightarrow ^{19}_{9}\mathrm{F}$$

Because ^{18}F and ^{19}F are isotopes of the same element with the same electron configuration, the chemical properties of these nuclei would be nearly identical.

(Choice A) Isotopes of the same element have the same electron configuration and exhibit nearly identical chemical properties.

(Choices B and D) If an ^{18}O nucleus absorbs an additional proton, its atomic number would increase from 8 to 9. Any atomic nucleus with 9 protons corresponds to the element fluorine.

Educational objective:
Isotopes are atoms of the same element that have the same number of protons but a different number of neutrons in the nucleus. Isotopes of the same element have nearly identical chemical properties but differ in their physical properties.

Time spent:0 sec QID:402458